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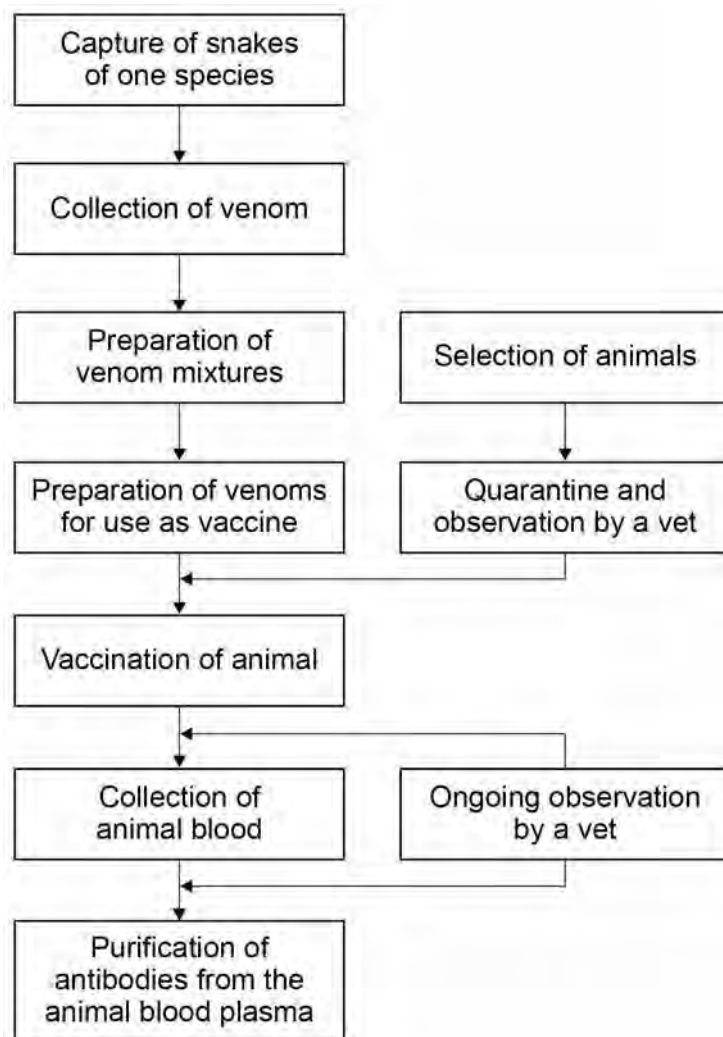
When a person is bitten by a venomous snake, the snake injects a toxin into the person. Antivenom is injected as treatment. Antivenom contains antibodies against the snake toxin. This treatment is an example of passive immunity.

Explain how the treatment with antivenom works and why it is essential to use passive immunity, rather than active immunity.

[2 marks]

Figure 8 shows a procedure used to produce antivenom.

Figure 8



0 7 . 2 A mixture of venoms from several snakes of the same species is used.

Suggest why.

[2 marks]

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0 7 . 3

Horses or rabbits can be used to produce antivenoms.

When taking blood to extract antibody, 13 cm^3 of blood is collected per kg of the animal's body mass.

The mean mass of the horses used is 350 kg and the mean mass of the rabbits used is 2 kg

Using only this information, suggest which animal would be better for the production of antivenoms.

Use a calculation to support your answer.

[2 marks]

0 7 . 4

During the procedure shown in **Figure 8** the animals are under ongoing observation by a vet.

Suggest **one** reason why.

[1 mark]
